Row Reduction and Row Echelon Forms

Recall from last week:

- A **linear equation** in the variable $x_1, \ldots, x_n$ is an equation that can be put in the following form
$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$
where b and the coefficients $a_1, \ldots, a_n$ are real or complex numbers. A **system of linear equations** is a collection of one or more linear equations involving the same variables. A **solution** to the linear system is a list of $(s_1, \ldots, s_n)$ of numbers that are solutions to each of the linear equations in the system.
- A linear system is **consistent** if there is a solution and **inconsistent** if there is no solution.
- A linear system has either no solution, exactly one solution, or infinitely many solutions.
- The following elementary row operations preserve solutions.
  1. Replace one row by the sum of itself and a multiple of another row.
  2. Interchange two rows.
  3. Multiply all entries in a row by a non-zero constant.
- Two matrices are called **row equivalent** if there is a sequence of elementary row operations changing one matrix to another.
- If the augmented matrices of two linear systems are row equivalent, then they both have the same solution set.
- Proof Techniques
  1. Quantifiers and counterexamples.
  2. Direct Proofs.
  3. Proof by cases.
  4. Proof by contrapositive.

Definition 1. *A matrix is in* **row echelon form** *if:*

1. *All rows consisting of only zeros are at the bottom.*
2. *The leading coefficient (also called the pivot) of a non-zero row is always strictly to the right of the leading coefficient of the row above it.*

1.

(a) Give an example of a matrix in echelon form.

(b) Give an example of a matrix not in echelon form.

Definition 2. *A matrix in row echelon form is in* **reduced row echelon form** *if*

(a) The leading entries are all 1.

(b) Each leading 1 is the only non-zero entry in its column.

(c) Give an example in reduced row echelon form.

Fact: Each matrix is row equivalent to one and only one reduced echelon matrix.

Definition 3. *A* **pivot position** *in a matrix A is a location in A that corresponds to a leading 1 in the reduced row echelon form. A* **pivot column** *is a column that has a pivot position.*

Row Reduction Algorithm (Gaussian Elimination).

(a) The leftmost non-zero column is a pivot column.

(b) Select a non-zero entry in the pivot column as pivot. Interchange rows to move this entry to the top.

(c) Use elementary row operations to create zeros below the pivot.

(d) Apply (a)-(c) to the submatrix obtained by removing pivot row and column. Repeat until the matrix is in row echelon form.

(e) Beginning with rightmost pivot and working upwards and to the left, create 0's above each pivot. Scale pivot positions entries to be 1.

2. Row-reduce the following matrix:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$$

3. Use row-reduction to solve the linear system

$$\begin{cases} x_1 + 3x_2 + 4x_3 = 7 \\ 3x_1 + 9x_2 + 7x_3 = 6 \end{cases}$$

Definition 4. *The variables corresponding to the pivot columns in an augmented matrix are called* **basic** *and the others are called* **free**.

Facts: A linear system is consistent if and only if the rightmost column in the augmented matrix is not a pivot column. If the system is consistent, then it has a unique solution when there are no free variables and infinite solutions when there is one or more free variables.

4. Find the general solution of the augmented matrix :

$$\left(\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right)$$

Definition 5. *A matrix with 1 column is called a* **column vector**. *A matrix with one row is called a* **row vector**.

We can add vectors of the same size by adding the corresponding components; namely, if $\vec{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, then $\vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}$. Similarly, we can scale a vector by a real number c, by multiplying all the components by c; namely, $c\vec{u} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}$.

Definition 6. *Given $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$ and scalars $c_1, \ldots c_p$, the vector*

$$\vec{y} = c_1\vec{v}_1 + \ldots c_p\vec{v}_p$$

is called the **linear combination** *of $\vec{v}_1, \ldots, \vec{v}_p$ with weights $c_1, \ldots, c_p$.*

5. Suppose $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$; and $\vec{w} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$. Can $\vec{w}$ be expressed as a linear combination of $\vec{u}$ and $\vec{v}$?

6. Suppose $\vec{a} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$; $\vec{b} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$; $\vec{c} = \begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix}$; and $\vec{d} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$. Can $\vec{d}$ be expressed as a linear combination of $\vec{a}$, $\vec{b}$, and $\vec{c}$?

A vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

has the same solutions as the augment matrix

$$[\vec{a}_1\ \vec{a}_2\ \ldots\ \vec{a}_n \,|\, \vec{b}].$$

Hence $\vec{b}$ is a linear combination of $\vec{a}_1, \ldots, \vec{a}_n$ if and only if there is a solution to this matrix.

Definition 7. *If $\vec{v}_1, \ldots, \vec{v}_p \in \mathbb{R}^n$, then the set of all linear combinations of $\vec{v}_1, \ldots, \vec{v}_p$ is called the* **span** *of $\vec{v}_1, \ldots, \vec{v}_p$. We write this set as Span$\{\vec{v}_1, \ldots, \vec{v}_p\}$.*

7.

(a) Suppose $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$; and $\vec{b} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$. Is $\vec{b}$ in Span$\{\vec{u}, \vec{v}\}$?

(b) Does there exist $\vec{b}$ not in Span$\{\vec{u}, \vec{v}\}$?

Row Echelon Form and Vector Equations – Answers / Solutions

1. (There are many examples for each of these.)

(a)

$$\begin{bmatrix} 0 & 1 & 10 & \pi \\ 0 & 0 & 0 & \ln(2) \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 0 & 1 & 10 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. Follow the algorithm:

The left column is nonzero, so it is a pivot column. The top entry 1 is not zero, so we use this for our pivot. Then we create zeroes below the pivot:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{R_2+(-2)R_1\to R_2} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 3 & 6 & 9 & 12 \end{bmatrix} \xrightarrow{R_3+(-3)R_1\to R_3} \begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & -3 & -12 \end{bmatrix}$$

Now we focus on the submatrix

$$\begin{bmatrix} 0 & -2 & -8 \\ 0 & -3 & -12 \end{bmatrix}$$

The leftmost nonzero column is the second column, so it is a pivot column. The top entry -2 is not zero, so we use this for our pivot. Then we create zeroes below the pivot:

$$\begin{bmatrix} 0 & -2 & -8 \\ 0 & -3 & -12 \end{bmatrix} \xrightarrow{R_2+(-\frac{3}{2})R_1\to R_2} \begin{bmatrix} 0 & -2 & -8 \\ 0 & 0 & 0 \end{bmatrix}$$

Putting this back into the original matrix:

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now all nonzero rows have been modified, so we need to zero entries above pivots starting from the right, and then scale the pivots to be 1.

$$\begin{bmatrix} 1 & 2 & 4 & 8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1+2R_2\to R_2} \begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{(-\frac{1}{2})R_2\to R_2} \begin{bmatrix} 1 & 2 & 0 & -8 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. Write this as an augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array}\right]$$

and follow the algorithm again. The left column is nonzero, so it is a pivot column. The top entry 1 is not zero, so we use this for our pivot. Then we create zeroes below the pivot.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{array}\right] \xrightarrow{R_2+(-3)R_1 \to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right]$$

Now all nonzero rows have been modified, so we need to zero entries above pivots starting from the right, and then scale the pivots to be 1.

$$\left[\begin{array}{ccc|c} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{array}\right] \xrightarrow{R_1+\frac{4}{5}R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & -5 & -15 \end{array}\right] \xrightarrow{\frac{-1}{5}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Translating back to equations:

$$\begin{cases} x_1 + 3x_2 = -5 \\ x_3 = 3 \end{cases}$$

So there are infinitely many solutions.

4. Follow the algorithm again. The left column is nonzero, so it is a pivot column. The top entry 3 is not zero, so we use this for our pivot. Then we create zeroes below the pivot.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 9 & -6 & 12 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right] \xrightarrow{R_2+(-3)R_1 \to R_2} \left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 6 & -4 & 8 & 0 \end{array}\right] \xrightarrow{R_3+(-2)R_1 \to R_3} \left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Now all nonzero rows have been modified, so we need to zero entries above pivots starting from the right, and then scale the pivots to be 1.

$$\left[\begin{array}{ccc|c} 3 & -2 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{3}R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{4}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

So this system is consistent with one basic variable (x_1) and two free variables (x_2, x_3). General solutions to this system have the form:

$$x_2 \begin{bmatrix} \frac{2}{3} \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -\frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$

5. No. Since $\vec{v} = 2\vec{u}$, then for any scalars c_u, c_v, the entries of $c_u\vec{u} + c_v\vec{v}$ are equal:

$$c_u\vec{u} + c_v\vec{v} = c_u\vec{u} + 2c_v\vec{u} = (c_u + 2c_v)\vec{u}$$

6. We could try the same approach as the previous problem, but when the vectors get larger and more complicated, it can be difficult to analyze them all. Let's use a more systematic approach, and change this to a row reduction problem.

We are asking whether there are scalars c_a, c_b, c_c such that $c_a\vec{a}+c_b\vec{b}+c_c\vec{c}=\vec{d}$. This is a system of linear equations with the variables c_a, c_b, c_c:

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -1 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right]$$

Now we follow the row reduction algorithm. The leftmost column is nonzero, and the top entry 1 is nonzero, so this is our pivot.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -1 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array}\right] \xrightarrow{R_2+R_1\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & 8 & 6 \end{array}\right]$$

Now the second column is nonzero and the leading entry of the second row is 1, so this is our second pivot.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & 8 & 6 \end{array}\right] \xrightarrow{R_3+(-2)R_2\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 10 & 4 \end{array}\right]$$

Now we create zeroes above pivot positions, and scale the pivot positions to be 1.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 10 & 4 \end{array}\right] \xrightarrow{\frac{1}{10}R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & \frac{2}{5} \end{array}\right] \xrightarrow{R_2+R_3\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 0 & \frac{7}{5} \\ 0 & 0 & 1 & \frac{2}{5} \end{array}\right]$$

$$\xrightarrow{R_1+(-5)R_3\to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & \frac{7}{5} \\ 0 & 0 & 1 & \frac{2}{5} \end{array}\right]$$

So we have our solution $c_a=0, c_b=\frac{7}{5}, c_c=\frac{2}{5}$. Just to be safe, let's check that this actually works:

$$0\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \frac{7}{5}\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + \frac{2}{5}\begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} = \vec{d}$$

7. (a) Row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 3 & -1 \\ 2 & 4 & 4 \end{array}\right] \xrightarrow{R_2+(-2)R_1\to R_2} \left[\begin{array}{cc|c} 1 & 3 & -1 \\ 0 & -2 & 6 \end{array}\right] \xrightarrow{-\frac{1}{2}R_2\to R_2} \left[\begin{array}{cc|c} 1 & 3 & -1 \\ 0 & 1 & -3 \end{array}\right]$$

$$\xrightarrow{R_1+(-3)R_2\to R_1} \left[\begin{array}{cc|c} 1 & 0 & 8 \\ 0 & 1 & -3 \end{array}\right]$$

so $\vec{b}\in \text{span}\{\vec{u},\vec{v}\}$ (Check: $8\begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-3)\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \vec{b}$).

(b) If $\vec{b} \notin \mathbb{R}^2$, then $\vec{b}$ is not in the span of $\vec{u}$ and $\vec{v}$. This is a somewhat silly answer to this question. Now assume $\vec{b} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$. Then we row reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 3 & x \\ 2 & 4 & y \end{array}\right] \xrightarrow{R_2+(-2)R_1 \to R_2} \left[\begin{array}{cc|c} 1 & 3 & x \\ 0 & -2 & y-2x \end{array}\right] \xrightarrow{-\frac{1}{2}R_2 \to R_2} \left[\begin{array}{cc|c} 1 & 3 & x \\ 0 & 1 & \frac{2x-y}{2} \end{array}\right]$$

$$\xrightarrow{R_1+(-3)R_2 \to R_1} \left[\begin{array}{cc|c} 1 & 0 & -2x+\frac{3}{2}y \\ 0 & 1 & \frac{2x-y}{2} \end{array}\right]$$

so $\vec{b} \in \operatorname{span}\{\vec{u}, \vec{v}\}$. (Check $(-2x + \frac{3}{2}y)\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2x-y}{2}\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \vec{b}$.)